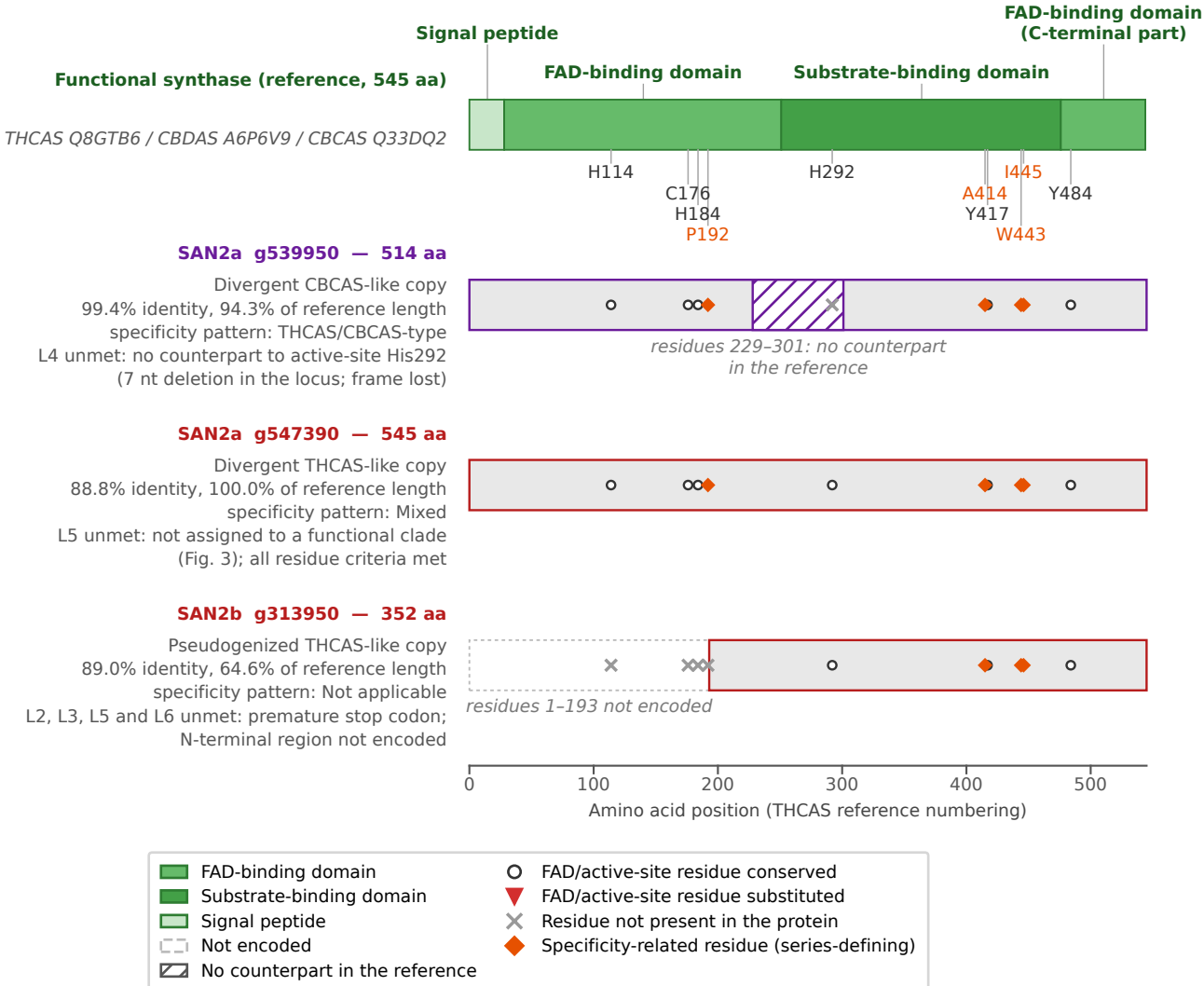
Signal peptide
FAD-binding domain
(C-terminal part)
Functional synthase (reference, 545 aa)
FAD-binding domain
Substrate-binding domain
THCAS Q8GTB6 / CBDAS A6P6V9 / CBCAS Q33DQ2
H114
C176
H184
P192
H292
A414
Y417
I445
W443
Y484
SAN2a g539950 — 514 aa
Divergent CBCAS-like copy
99.4% identity, 94.3% of reference length
specificity pattern: THCAS/CBCAS-type
L4 unmet: no counterpart to active-site His292
(7 nt deletion in the locus; frame lost)
residues 229–301: no counterpart
in the reference
SAN2a g547390 — 545 aa
Divergent THCAS-like copy
88.8% identity, 100.0% of reference length
specificity pattern: Mixed
L5 unmet: not assigned to a functional clade
(Fig. 3); all residue criteria met
SAN2b g313950 — 352 aa
Pseudogenized THCAS-like copy
89.0% identity, 64.6% of reference length
specificity pattern: Not applicable
L2, L3, L5 and L6 unmet: premature stop codon;
N-terminal region not encoded
residues 1–193 not encoded
0
100
200
300
400
500
Amino acid position (THCAS reference numbering)
FAD-binding domain
Substrate-binding domain
Signal peptide
Not encoded
No counterpart in the reference
FAD/active-site residue conserved
FAD/active-site residue substituted
Residue not present in the protein
Specificity-related residue (series-defining)